

SINPO, Korea, Democratic People's Republic ofWMO: **470460**Lat: **40.033N**Lon: **128.183E**Elev: **62**StdP: **14.66**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	14.2	-7.8	3.6	17.8	-3.5	4.6	19.4	22.8	25.7	20.3	32.7	2.6	270	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	83.9	76.1	81.9	75.2	79.9	73.9	77.7	82.4	76.4	80.7	75.1	78.9	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	137.1	81.2	75.0	131.4	79.6	73.7	125.9	78.3	41.3	82.7	40.0	80.8	38.8	79.0	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.4	14.3	4.7	87.8	3.4	1.9	2.2	89.2	0.3	90.3	-1.6	91.4	-4.1	92.8		
(5)			3.5	80.4	3.2	2.2	1.2	82.0	-0.7	83.3	-2.5	84.5	-4.8	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	27.1	29.8	38.4	48.8	57.6	65.5	72.6	74.1	66.1	55.8	42.6	31.5
	DBStd	17.05	5.69	5.50	5.72	5.73	5.41	4.32	4.89	3.70	4.03	5.19	6.69	6.48
	HDD50	2561	711	566	363	91	7	0	0	0	0	13	235	575
	HDD65	5792	1176	986	825	488	236	46	3	0	34	285	673	1040
	CDD50	2900	0	0	2	54	241	464	702	748	483	194	12	0
	CDD65	656	0	0	0	0	5	60	240	283	67	1	0	0
	CDH74	2962	0	0	0	0	30	132	1197	1458	144	1	0	0
	CDH80	412	0	0	0	0	2	13	205	186	6	0	0	0
Wind	WSAvg	4.7	4.0	4.2	5.4	5.7	5.2	4.6	4.8	4.2	4.1	4.5	4.9	4.2
Precipitation	PrecAvg	28.3	0.6	0.7	0.7	1.8	2.5	3.0	6.6	5.6	3.5	1.6	1.4	0.7
	PrecMax	48.9	2.4	2.4	3.1	8.1	5.6	8.2	14.5	15.3	9.4	3.3	4.0	3.4
	PrecMin	15.1	0.2	0.0	0.1	0.2	0.2	0.6	0.5	0.7	0.9	0.0	0.0	0.0
	PrecStd	7.2	0.4	0.7	0.6	1.5	1.5	2.0	3.4	3.5	2.4	1.0	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	47.4	58.5	68.9	77.4	82.0	87.0	86.6	81.1	71.8	63.0	49.5
		MCWB	38.0	39.7	48.0	55.2	60.8	68.9	77.1	78.4	71.8	62.8	53.4	42.8
	2%	DB	40.8	43.7	53.8	65.0	72.8	77.6	84.4	83.9	77.9	69.1	59.2	46.2
		MCWB	35.4	38.0	45.2	53.0	59.6	67.2	76.4	77.0	69.7	61.2	51.7	40.7
	5%	DB	38.7	40.7	50.7	61.8	69.3	74.7	82.3	82.1	75.5	67.0	56.5	43.7
		MCWB	33.8	35.8	43.5	51.2	58.1	67.0	75.7	75.9	67.9	59.7	49.6	38.7
	10%	DB	36.5	38.4	47.5	58.8	66.5	72.5	80.1	80.2	73.5	65.0	53.6	41.2
		MCWB	32.1	34.0	41.4	49.9	57.4	66.4	74.5	74.7	66.9	58.3	47.6	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	41.2	49.5	57.4	64.4	72.0	79.5	79.8	74.4	65.7	55.4	44.5
		MCDB	43.3	46.2	56.5	66.1	71.2	77.5	84.2	84.6	79.5	70.4	59.8	47.7
	2%	WB	36.3	38.6	46.3	54.6	62.0	70.0	77.7	78.1	71.3	62.9	53.0	41.6
		MCDB	39.8	42.7	51.9	62.3	69.6	74.4	82.8	82.6	75.8	67.1	58.5	45.2
	5%	WB	34.6	36.4	44.1	52.4	60.0	68.6	76.3	76.9	69.6	61.2	50.5	39.3
		MCDB	37.9	40.2	49.6	59.9	66.6	72.9	81.2	81.0	73.7	65.7	55.6	43.0
	10%	WB	32.7	34.5	42.1	50.3	58.4	67.1	75.0	75.7	68.3	59.3	48.2	37.0
		MCDB	35.9	38.0	46.9	57.6	64.5	71.3	79.4	79.3	72.4	63.9	52.7	40.7
Mean Daily Temperature Range	5% DB	MDBR	14.5	13.1	12.2	13.7	12.2	9.0	7.9	8.9	12.4	14.5	13.7	14.1
		MCDBR	15.7	14.7	16.3	18.6	18.5	14.8	12.1	11.3	14.5	16.1	16.8	15.3
		MCWBR	12.6	11.0	10.6	10.3	9.2	7.9	6.9	6.6	8.9	10.9	12.1	11.9
	5% WB	MCDBR	14.4	13.6	14.7	16.3	15.2	10.8	10.8	9.5	11.1	14.2	15.3	14.1
		MCWBR	12.0	10.5	10.4	10.3	8.9	6.7	6.8	6.2	7.6	10.4	11.7	11.7
Clear-Sky Solar Irradiance	taub		0.373	0.428	0.501	0.569	0.594	0.634	0.588	0.522	0.465	0.463	0.420	0.372
	taud		2.088	1.971	1.834	1.736	1.749	1.738	1.910	2.082	2.161	2.023	2.055	2.119
	Ebn at Noon		238	240	235	229	227	218	227	239	246	226	218	227
	Edh at Noon		36	47	60	71	71	72	60	49	43	44	37	33
All-Sky Solar Radiation	RadAvg		770	1061	1409	1636	1739	1624	1435	1425	1326	1092	762	672
	RadStd		52	79	100	126	177	157	184	127	134	93	110	55